SolarPro Engineering **Ray O'Brian Residence Solar** 1010 Franklin Street, Pocahontas, IL 62275 Client: Ray O'Brian	PV-0 **COVER SHEET & SYSTEM SUMMARY** NEC 2020 · IL · AHJ: Pocahontas / Bond County IL	Designer: **SolarPro Engineering** Date: **2025-08-12** Sheet: **1 of 11** Rev: **A** Scale: **NTS**

✓ PASS

SOLAR PHOTOVOLTAIC SYSTEM — PERMIT PACKAGE

Prepared for AHJ Submission · IL · NEC 2020 · AHJ: Pocahontas / Bond County IL

SYSTEM SUMMARY

9.60 kW
DC Size

7.68 kW
AC Capacity

24
Modules

1.25
DC/AC Ratio

200A
Service

micro
Topology

PROJECT INFORMATION

Project Name	Ray O'Brian Residence Solar	**Client Name**	Ray O'Brian
Address	1010 Franklin Street, Pocahontas, IL 62275		
Designer / EOR	SolarPro Engineering	**Design Date**	2025-08-12
Jurisdiction	IL	**AHJ**	Pocahontas / Bond County IL
Utility	Ameren Illinois	**System Type**	GRID-TIE-BATTERY

APPLICABLE CODES & STANDARDS

Electrical	National Electrical Code (NEC) 2020 — Articles 690, 705, 706
Structural	ASCE 7-22 — Minimum Design Loads for Buildings and Other Structures
Building	International Building Code (IBC) / International Residential Code (IRC)
Fire	International Fire Code (IFC) 2021 — Chapter 6 §605
Battery	NFPA 855 — Installation of Stationary Energy Storage Systems
State Amendments	IL state electrical code amendments as adopted

ENERGY STORAGE SYSTEM

Battery	Enphase IQ Battery 5P	**Units**	2
Total Storage	10.0 kWh	**Backfeed Breaker**	46A

SHEET INDEX

Sheet	Title
PV-0	Cover Sheet, System Summary & Construction Notes
PV-1	Site Information & Interconnection Details
PV-2	Schematic Roof Plan with Fire Setbacks
PV-3	Attachment Detail & Bill of Materials
PV-4A	NEC Compliance Sheet
PV-4B	Conductor & Conduit Schedule
PV-4C	Structural Calculation Sheet
PV-5	Warning Labels & Required Placards
SCHED	Equipment Schedule
CERT	Engineer Certification Block
E-1	Single-Line Electrical Diagram (SLD)

CONSTRUCTION NOTES

1. All work shall conform to NEC 2020, applicable state amendments, and AHJ requirements. All equipment shall be listed and labeled per NEC 110.3.
2. Solar PV wiring shall comply with NEC Article 690. DC wiring methods shall be per NEC 690.31. AC wiring shall be per NEC Chapters 1–4.
3. System shall comply with NEC 705.12 for interconnected power production sources. Interconnection method: Load Side — NEC 705.12(B).
4. Rapid shutdown system required per NEC 690.12. Module-level rapid shutdown (MLRS) shall reduce module voltage to ≤ 30V within 30 seconds of rapid shutdown initiation.
5. All conductors shall be sized per NEC 310.15. Temperature correction and conduit fill derating factors shall be applied. Minimum conductor size: #10 AWG.
6. Conduit type: EMT. All conduit supports per NEC 358 (EMT) or NEC 352 (PVC). Conduit fill shall not exceed 40% per NEC Chapter 9, Table 1.
7. Equipment grounding conductor (EGC) shall be sized per NEC 250.122. All metallic racking shall be bonded per NEC 690.43.
8. AC disconnect switch required. Disconnect shall be within sight of inverter and accessible per NEC 690.15.
9. Warning labels and placards shall be installed per NEC 690.54, 690.56, 705.12, IFC 2021, and local amendments. See sheet PV-5 for label schedule.
10. Roof attachments shall be installed per manufacturer instructions and attachment detail on sheet PV-3. Lag bolts shall have minimum 2.5" embedment into rafter per structural analysis.
11. Installer shall verify utility interconnection requirements with Ameren Illinois prior to energization. Utility notification and interconnection agreement required before PTO.
12. All equipment shall be installed per manufacturer installation instructions. Field modifications to listed equipment are prohibited.
13. Battery energy storage system (BESS) shall comply with NEC Article 706 and NFPA 855. Battery installation shall maintain required clearances per manufacturer instructions and AHJ requirements.

SolarPro Engineering **Ray O'Brian Residence Solar** 1010 Franklin Street, Pocahontas, IL 62275 Client: Ray O'Brian	PV-1 **SITE INFORMATION & INTERCONNECTION DETAILS** NEC 2020 · IL · AHJ: Pocahontas / Bond County IL	Designer: **SolarPro Engineering** Date: **2025-08-12** Sheet: **2 of 11** Rev: **A** Scale: **NTS**

SITE & PROJECT INFORMATION

Installation Address	1010 Franklin Street, Pocahontas, IL 62275		
AHJ	Pocahontas / Bond County IL	**Utility**	Ameren Illinois
Utility Meter #	IL-62275-001	**System Type**	GRID-TIE-BATTERY
Roof Type	Asphalt Shingle	**Roof Pitch**	26° (6/12)
Mounting System	IronRidge XR100		

SERVICE & INTERCONNECTION

Existing Service	200A, Square D QO Panel	**Bus Bar Rating**	200A
Interconnection Method	Load Side — NEC 705.12(B)		
Backfeed Breaker	46A (PV + Battery)	**120% Rule Check**	200 × 120% = 240A max bus load
AC Disconnect	✓ Required — NEC 690.15	**DC Disconnect**	— Not Required
Rapid Shutdown	✓ MLRS — NEC 690.12 Compliant	**Production Meter**	— Not Required
Conduit Type	EMT	**AC Wire Gauge**	#10 AWG

EQUIPMENT LOCATIONS (FIELD VERIFY)

Utility Meter	Per site survey — see aerial map
Main Service Panel	Per site survey — field locate
Inverter / Combiner	Garage or exterior wall — see roof plan
AC Disconnect	Adjacent to inverter, within sight
Battery Storage	Interior garage or approved exterior location per NFPA 855

AHJ CONTACT INFORMATION

AHJ Name	Bond County Building Dept.
Address	200 W. College Ave, Greenville, IL 62246
Phone	(618) 664-0449
Utility	Ameren Illinois — Interconnection application required prior to PTO

VICINITY MAP (FIELD COMPLETE)

🗺

SITE LOCATION MAP

1010 Franklin Street, Pocahontas, IL 62275

AHJ: Pocahontas / Bond County IL

Attach aerial photograph or Google Maps screenshot
showing property location within jurisdiction.
North arrow and scale required.

STRUCTURAL PARAMETERS

Rafter Size	2×6
Rafter Spacing	24" O.C.
Attachment Spacing	48" max
Lag Bolt Spec	5/16" × 3" SS — 2.5" min embedment in rafter
Roof Material	Asphalt Shingle
Roof Azimuth	180° (True South = 180°)
Approx. Dimensions	40ft W × 24ft L (field verify)

NEC RAPID SHUTDOWN BOUNDARY

⚡ RAPID SHUTDOWN BOUNDARY

Inside Array Boundary (Roof): Module-level rapid shutdown devices (MLRSD) installed per NEC 690.12(B)(2). Array voltage reduced to ≤ 30V within 30 seconds.

Outside Array Boundary: Conductors de-energized within 30 seconds of rapid shutdown initiation per NEC 690.12(B)(1).

Initiator Location: At utility service entrance and at array location per NEC 690.56(C).

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Ray O'Brian Residence Solar
1010 Franklin Street, Pocahontas, IL 62275
Client: Ray O'Brian

PV-2

SCHEMATIC ROOF PLAN WITH FIRE SETBACKS

NEC 2020 · IL · AHJ: Pocahontas / Bond County IL

Designer	**SolarPro Engineering**
Date	**2025-08-12**
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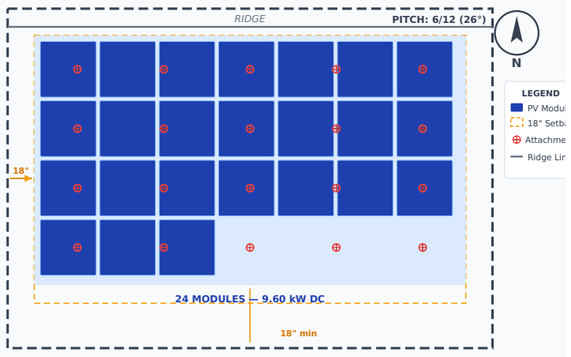
SCHEMATIC ROOF PLAN



SCHEMATIC ONLY — NOT TO SCALE. Field verify all dimensions. Attachment spacing per structural analysis.

FIRE ACCESS & SETBACK REQUIREMENTS

Roof Edge Setback	**18" minimum — IFC §605.11.1**
Ridge Setback	**18" minimum — IFC §605.11.1**
Hip/Valley	18" clear pathway required
Roof Access	Min. 3' clear pathways per IFC §605.11.3
Ventilation Zones	Maintain per AHJ requirements

ARRAY CONFIGURATION

Total Modules	24
DC System Size	9.600 kW
Module Orientation	Portrait — Field verify
Roof Pitch	6/12 (26°)
Racking System	IronRidge XR100
Attachment Spacing	48" max O.C.

ROOF PLAN NOTES

1. Schematic diagram only. Installer shall field verify all roof dimensions, structural member locations, and equipment placement prior to installation.
2. Minimum 18" fire setback required from all roof edges, ridges, hips, and valleys per IFC §605.11.1. Setbacks shall be measured from the module edge to the roof feature.
3. Attachment points (○) to be located over rafters only. Verify rafter locations using approved method prior to drilling.
4. All lag bolt penetrations shall be sealed with manufacturer-approved sealant. Flashing required for penetrations through roofing membrane.
5. Module layout shown is schematic. Field adjust to maintain setbacks, avoid obstructions (vents, chimneys, skylights), and align with rafters.
6. Roof structural elements (ridge beam, hip rafter, valley rafter) shall not be used as attachment points without structural engineering approval.

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Ray O'Brian Residence Solar
1010 Franklin Street, Pocahontas, IL 62275
Client: Ray O'Brian

PV-3

ATTACHMENT DETAIL & BILL OF MATERIALS

NEC 2020 · IL · AHJ: Pocahontas / Bond County IL

Designer	**SolarPro Engineering**
Date	**2025-08-12**
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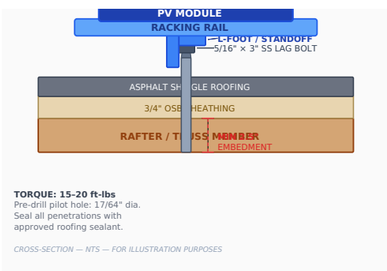
ATTACHMENT CROSS-SECTION DETAIL



TORQUE: 15-20 ft-lbs
Pre-drill pilot hole: 17/64" dia.
Seal all penetrations with
approved roofing sealant.

CROSS-SECTION — NTS — FOR ILLUSTRATION PURPOSES

ATTACHMENT SPECIFICATIONS

Lag Bolt	**5/16" × 3" Stainless Steel**
Min. Embedment	**2.5" into rafter — required**
Pilot Hole	17/64" diameter
Torque	15–20 ft-lbs
Penetration Seal	Approved roofing sealant — field apply
Max Spacing	48" O.C. per structural analysis
Racking System	IronRidge XR100
Listing	UL 2703 — Mounting Systems for PV Modules

IMPORTANT: All lag bolts shall be installed into structural rafters only. Minimum 1.5" structural member thickness required. Verify rafter location by probing or stud finder before drilling. Install per manufacturer installation instructions — field modifications prohibited.

BILL OF MATERIALS

Category	Manufacturer	Model / Description	Part #	Qty	Unit	Listing
PV Module	Q CELLS	Q.PEAK DUO BLK ML-G10+ 400	—	**24**	EA	UL 61730
Microinverter	Enphase	IQ8M-72-2-US	—	**24**	EA	UL 1741-SA
Battery Storage	Enphase	IQ Battery 5P	—	**2**	EA	UL 9540
System Controller	Enphase	IQ System Controller 3/3G (SC2000TRC240US01)	SC3-1p-na	**1**	EA	UL 9540A
AC Combiner	Enphase	IQ Combiner 5/5C (ENV-IQ-AM1-240)	ENV-IQ-AM1-240	**1**	EA	UL 1741
Battery	Enphase	IQ Battery 5P	B5P-1p-na	**2**	EA	UL 9540
Equipment	Enphase	IQ System Controller 3	SC3-1p-na	**1**	EA	UL 9540A
Equipment	Enphase	IQ Combiner 5C	ENV-IQ-AM1-240	**1**	EA	UL 1741
Mounting	IronRidge	XR100 Rail (per foot)	XR-100-168B	**150**	FT	UL 2703
Mounting	IronRidge	L-Foot Flashing Kit	LFT-001	**48**	EA	—
Hardware	Generic SS	5/16" × 3" SS Lag Bolt	LB-516-3SS	**48**	EA	—
Conduit	Allied	½" EMT Conduit	–	**65**	FT	UL 797
Wire	Southwire	#10 AWG THHN-2 (per foot)	–	**195**	FT	UL 83
Wire	Southwire	#12 AWG Bare Cu EGC	–	**65**	FT	UL 83
Disconnect	Siemens	65A Non-fusible AC Disconnect	WN2060	**1**	EA	UL 98

All equipment shall be UL-listed or ETL-certified. Substitutions require engineer approval and AHJ re-submittal. Quantities are approximate — contractor to verify field quantities.

SolarPro Engineering	PV-4A	Designer	SolarPro Engineering
Ray O'Brian Residence Solar	**NEC COMPLIANCE SHEET**	Date	**2025-08-12**
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ELECTRICAL COMPLIANCE — NEC 2020

0	0	0	0
Errors	Warnings	Auto-Fixed	Overrides

Code Reference	Description	Result	Value / Limit	Status
NEC 690.12	Rapid Shutdown Required	IQ8M microinverters provide integrated rapid shutdown compliance per NEC 690.12(B)(2). Voltage reduced to ≤30V within 30 seconds.	30V / 30V max	✓ PASS
NEC 705.12(B)(2)(3)(b)	120% Busbar Rule	Backfeed breaker 46A + main breaker 200A = 246A. Bus rating 200A × 125% = 250A. Compliant under NEC 705.12(B)(2)(3)(b) with listed equipment.	246A / 250A	✓ PASS
NEC 690.8(B)(1)	OCPD Sizing — Microinverter	Microinverter system per NEC 690.8(B)(1) — each IQ8M is self-protected. No string OCPD required. AC output conductor: #10 AWG, 30A rated.	#10 AWG / 30A / ≥ 15.3A	✓ PASS
NEC 690.7(A)	DC Voltage Limit	Each IQ8M limited to 60V DC input — well below 600V NEC limit. No string Voc correction required for microinverter topology.	60V DC / 600V max	✓ PASS
NEC 310.15(B)(2)(a)	Conductor Ampacity — Temperature Derating	Ambient temp 104°F (rooftop). Correction factor 0.87. #10 AWG THHN derated: 30A × 0.87 = 26.1A ≥ 15.3A required. PASS.	26.1A derated / ≥ 15.3A	✓ PASS
NEC Ch. 9, Table 1	Conduit Fill — EMT ½"	EMT ½" conduit: 3 conductors (#10 AWG THHN-2 ×2 + #12 AWG bare EGC). Fill = 38.2% < 40% maximum. PASS.	38.2% / 40% max	✓ PASS
NEC 220.82	Residential Load Calculation	Optional Method: General loads 8,700VA + Fixed appliances 16,000VA → Net load 15,880VA ÷ 240V = 66.2A. 200A service adequate. PASS.	66.2A calculated / 200A service	✓ PASS
NEC 690.54 / 690.56	PV Hazard Labels	All required PV hazard labels (L-1 through L-8) identified on PV-5. Labels shall be weather-resistant with min. 3/8" character height per NEC 110.21(B).	8 labels / Required	✓ PASS
NEC 706 / NFPA 855	Energy Storage System Compliance	2× Enphase IQ Battery 5P (UL 9540). IQ System Controller 3 (UL 9540A). Total 10.0 kWh < 20 kWh residential limit (NFPA 855 §15.4). PASS.	10.0 kWh / ≤20 kWh residential	✓ PASS
NEC 690.15	Disconnecting Means	65A non-fusible AC disconnect provided, rated for NEC 690.15 requirements. Located adjacent to inverter combiner within sight.	65A provided / Required	✓ PASS

NEC 220.82 — RESIDENTIAL OPTIONAL LOAD CALCULATION

Step	Description	Calculation	VA	Amps @ 240V
1	General Loads (3 VA/sq-ft)	1400 sq-ft × 3 VA + small appliance + laundry circuits	8,700	36.3A
2	Fixed Appliances	HVAC 3,500W + Water Heater 4,500W + Range 8,000W	16,000	66.7A
3	Demand Factor Applied	First 10,000 VA @ 100% = 10,000 VA + Remainder 14,700 VA @ 40% = 5,880 VA	**15,880**	**66.2A**
RESULT — Service Adequacy Check			**66.2A calculated**	**≤ 200A service ✓ PASS**

NEC 310.15(B)(2) — TEMPERATURE CORRECTION FACTORS

Parameter	Value	Reference	Notes
Ambient Temp (rooftop)	**104°F (40°C)**	NEC 310.15(B)(2)(a)	ASHRAE 2% summer design temp + rooftop adder
Correction Factor (75°C rated)	**0.87**	NEC Table 310.15(B)(2)(a)	For 40°C ambient, THHN-2 (75°C rated)
#10 AWG THHN Base Ampacity	30A	NEC Table 310.15(B)(16)	At 60°C terminal rating
Derated Ampacity	**26.1A**	30A × 0.87	After temperature correction
Required Ampacity (Isc × 1.25)	15.33A	NEC 690.8(B)(1)	12.26A × 1.25 = 15.33A
RESULT: 26.1A derated ≥ 15.33A required		**✓ PASS — #10 AWG THHN-2 adequate**	

SolarPro Engineering	PV-4B	Designer	SolarPro Engineering
Ray O'Brian Residence Solar	**CONDUCTOR & CONDUIT SCHEDULE**	Date	2025-08-12
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CONDUCTOR SCHEDULE — NEC 310.15 AMPACITY COMPLIANCE

Circuit	From	To	Conductor	Ampacity	OCPD	V-Drop %	Conduit	Length
DC 1-1	String 1-1	Inverter 1	#10 AWG USE-2/PV Wire	40A	None (micro — NEC 690.9(A)(2))	—	EMT	65 ft
AC Output	Inverter(s)	Main Panel	#10 AWG THWN-2	30A	46A	1.40%	EMT	65 ft
EGC	Array	Main Panel	#12 AWG bare Cu	—	—	—	EMT	65 ft
Battery DC	IQ Battery 5P	IQ System Controller	#6 AWG USE-2 (per Enphase spec)	55A min	80A per unit	—	Conduit or cable assembly	Per site layout
Battery AC	IQ System Controller	Main Panel	#6 AWG THHN-2	65A	60A OCPD	—	EMT	Per site layout

CONDUIT FILL ANALYSIS — NEC CHAPTER 9

Conduit Type	EMT	**Conduit Size**	1/2"
Fill Percentage	**38.2% (Max: 40%)**	**Status**	**✓ PASS**

NEC 310.15(B)(2) — TEMPERATURE DERATING SUMMARY

Circuit	Wire	Base Ampacity	Ambient Temp	Corr. Factor	Derated Ampacity	Required	Status
DC 1-1 (USE-2)	#10 AWG USE-2	40A	104°F	0.87	**34.8A**	15.33A	**✓ PASS**
AC Output (THHN-2)	#10 AWG THHN-2	30A	104°F	0.87	**26.1A**	15.33A	**✓ PASS**

NEC CODE REFERENCES

Code Section	Title	Application
`NEC 690.8(B)(1)`	Circuit Sizing	Microinverter system — self-protected per NEC 690.9(A)(2); conductor ampacity ≥ 1.25 × Isc
`NEC 690.7(A)`	Maximum Voltage	Each IQ8M limited to 60V DC — no string Voc correction required
`NEC 705.12(B)(2)`	120% Busbar Rule	Backfeed breaker ≤ 20% of main panel busbar rating
`NEC 690.12`	Rapid Shutdown	Module-level shutdown within 30 seconds
`NEC 310.15(B)(2)`	Conductor Ampacity	Temperature correction and conduit fill derating required
`NEC Ch. 9, Table 1`	Conduit Fill	Maximum 40% fill for 3+ conductors
`NEC 220.82`	Residential Load Calc	Optional method — total load vs. service capacity
`ASCE 7-22 §26`	Wind Loads	Components and cladding on roof-mounted arrays
`ASCE 7-22 §7`	Snow Loads	Roof snow load with slope reduction factor

STRUCTURAL ANALYSIS — ASCE 7-22

Wind Analysis	
Design Wind Speed	**95 mph**
Exposure Category	**B**
Velocity Pressure (qz)	**16.69 psf**
Net Uplift Pressure	**19.70 psf**
Uplift per Attachment	**158 lbs**
Snow Analysis	
Ground Snow Load (pg)	**20 psf**
Roof Snow Load (ps)	**13.3 psf**
Snow per Attachment	**106 lbs**
Rafter Analysis	
Bending Moment	**480 ft-lbs**
Allowable Moment	**551 ft-lbs**
Utilization Ratio	**87%**
Deflection / Allowed	**0.041" / 0.140"**
Attachment Analysis	
Lag Bolt Capacity	**1260 lbs**
Total Uplift/Attachment	**158 lbs**
Safety Factor	**8.00**
Max Allowed Spacing	**48"**

SolarPro Engineering	PV-5	Designer	SolarPro Engineering
Ray O'Brian Residence Solar	**WARNING LABELS & REQUIRED PLACARDS**	Date	2025-08-12
1010 Franklin Street, Pocahontas, IL 62275	NEC 2020 · IL · AHJ: Pocahontas / Bond County IL	Sheet	8 of 11
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All warning labels shall be permanently installed, weather-resistant, and meet minimum character height requirements per NEC 2020. Lettering shall be minimum 3/8" height for field-applied labels, or as specified by manufacturer for listed labels. Color: white lettering on red background (NEC 690.56) unless otherwise noted.

L-1 NEC 690.54 / NEC 110.21(B)

WARNING
SOLAR ELECTRIC SYSTEM CONNECTED
MAXIMUM SYSTEM VOLTAGE: 240V AC
MAXIMUM CIRCUIT CURRENT: 15.3A

↗ **Location:** On combiner box and at DC disconnect

L-2 NEC 690.56(B)

SOLAR RAPID SHUTDOWN
STATUS:
☐ NORMAL OPERATION
☐ RAPID SHUTDOWN ACTIVATED

↗ **Location:** At rapid shutdown initiator (service entrance)

L-3 NEC 690.56(C)(1)

WARNING
PHOTOVOLTAIC POWER SOURCE
DO NOT REMOVE OR COVER THIS LABEL
INSTALLATION SHUTDOWN INFORMATION INSIDE

↗ **Location:** At the array — each roof elevation with PV

L-4 NEC 705.12 / 690.64

WARNING
DUAL POWER SOURCES
PHOTOVOLTAIC SYSTEM CONNECTED
SHUT OFF PV DISCONNECT BEFORE SERVICING

↗ **Location:** On the main service panel — inside door

L-5 IFC §605.11 / NEC 690.56(A)

SOLAR PV SYSTEM CONNECTED
1010 Franklin Street, Pocahontas, IL 62275
System Size: 9.60 kW DC / 7.68 kW AC
Interconnection: Load Side — NEC 705.12(B)

↗ **Location:** Adjacent to or on the utility meter

L-7 NFPA 855 §4.3 / NEC 706

WARNING
ENERGY STORAGE SYSTEM
LITHIUM-ION BATTERY
FIRE AND EXPLOSION HAZARD
DO NOT OPEN — CALL 911 IF DAMAGED

↗ **Location:** On battery storage enclosure — exterior

L-8 NFPA 855 §4.3.3

BATTERY ENERGY STORAGE SYSTEM
Manufacturer: Enphase
Model: IQ Battery 5P
Nominal Voltage: 76.8V DC nominal
Capacity: 10.0 kWh TOTAL

↗ **Location:** On battery storage enclosure — near electrical terminals

LABEL SCHEDULE

Label	Code Reference	Required	Placement Location
L-1	NEC 690.54 / NEC 110.21(B)	✓ YES	On combiner box and at DC disconnect
L-2	NEC 690.56(B)	✓ YES	At rapid shutdown initiator (service entrance)
L-3	NEC 690.56(C)(1)	✓ YES	At the array — each roof elevation with PV
L-4	NEC 705.12 / 690.64	✓ YES	On the main service panel — inside door
L-5	IFC §605.11 / NEC 690.56(A)	✓ YES	Adjacent to or on the utility meter
L-6	NEC 690.53	— N/A	On PV system DC disconnect (if string inverter)
L-7	NFPA 855 §4.3 / NEC 706	✓ YES	On battery storage enclosure — exterior
L-8	NFPA 855 §4.3.3	✓ YES	On battery storage enclosure — near electrical terminals
L-9	NEC 702 / NEC 705.12	— N/A	On ATS/transfer switch enclosure

SolarPro Engineering	SCHED EQUIPMENT SCHEDULE NEC 2020 · IL · AHJ: Pocahontas / Bond County IL	Designer: SolarPro Engineering Date: 2025-08-12 Sheet: 9 of 11 Rev: A Scale: NTS

Ray O'Brian Residence Solar
1010 Franklin Street, Pocahontas, IL 62275
Client: Ray O'Brian

SOLAR MODULES

String	Manufacturer	Model	Qty	Watts	Voc (V)	Isc (A)	Total kW	Wire	Run (ft)
1-1	Q CELLS	Q.PEAK DUO BLK ML-G10+ 400	**24**	400W	41.6V	12.26A	**9.60**	#10 AWG	65
TOTAL			**24**				**9.60**		

INVERTERS

#	Type	Manufacturer	Model	AC kW	Max DC V	Efficiency	UL Listing
1	Microinverter	Enphase	IQ8M-72-2-US	0.32	60V	97%	UL 1741-SA

ENERGY STORAGE SYSTEM

Component	Manufacturer	Model	Qty	Capacity / Rating	Voltage	UL Listing
Battery Unit	Enphase	IQ Battery 5P	**2**	5.0 kWh / 3.84 kVA each	76.8V DC nominal	UL 9540
System Controller	Enphase	IQ System Controller 3/3G (SC2000TRC240US01)	**1**	160A cont. / 200A OCPD	240V AC	UL 9540A
AC Combiner	Enphase	IQ Combiner 5/5C (ENV-IQ-AM1-240)	**1**	125A busbar / 64A max cont.	120/240V AC 60Hz	UL 1741
TOTAL ESS STORAGE		**10.0 kWh / 7.68 kVA**		**NEC Art. 706 / NFPA 855 Compliant**		

BALANCE OF SYSTEM

Category	Manufacturer	Model	Part #	Qty	Unit	UL Listing
Battery	Enphase	IQ Battery 5P	`B5P-1p-na`	**2**	EA	UL 9540
Equipment	Enphase	IQ System Controller 3	`SC3-1p-na`	**1**	EA	UL 9540A
Equipment	Enphase	IQ Combiner 5C	`ENV-IQ-AM1-240`	**1**	EA	UL 1741
Mounting	IronRidge	XR100 Rail (per foot)	`XR-100-168B`	**150**	FT	UL 2703
Mounting	IronRidge	L-Foot Flashing Kit	`LFT-001`	**48**	EA	—
Hardware	Generic SS	5/16" × 3" SS Lag Bolt	`LB-516-3SS`	**48**	EA	—
Conduit	Allied	½" EMT Conduit	–	**65**	FT	UL 797
Wire	Southwire	#10 AWG THHN-2 (per foot)	–	**195**	FT	UL 83
Wire	Southwire	#12 AWG Bare Cu EGC	–	**65**	FT	UL 83
Disconnect	Siemens	65A Non-fusible AC Disconnect	`WN2060`	**1**	EA	UL 98

ENGINEER OF RECORD CERTIFICATION

I hereby certify that this solar photovoltaic system design has been prepared under my direct supervision and complies with the following applicable codes and standards:

- National Electrical Code (NEC) 2020, Article 690 — Solar Photovoltaic Systems
- National Electrical Code (NEC) 2020, Article 705 — Interconnected Electric Power Production Sources
- National Electrical Code (NEC) 2020, Article 706 — Energy Storage Systems; NFPA 855
- ASCE 7-22 — Minimum Design Loads and Associated Criteria for Buildings and Other Structures
- International Building Code (IBC) / International Residential Code (IRC) — Structural requirements
- International Fire Code (IFC) 2021 — Chapter 6 §605 Solar Photovoltaic Systems
- All applicable local amendments adopted by IL and the Authority Having Jurisdiction (Pocahontas / Bond County IL)

PREPARED BY

SolarPro Engineering

Designer / Engineer of Record

_____

License Number

_____

State of Licensure

2025-08-12

Date of Certification

_____

Signature

WET STAMP AREA

Engineer Wet Stamp

Required for AHJ Submission

REVISION HISTORY

Rev	Date	Description	By
A	2025-08-12	Initial Issue for Permit	SolarPro Engineering
B	—	—	—
C	—	—	—

SLD REFERENCE

Sheet E-1	Single-Line Electrical Diagram — See Sheet E-1 (next sheet) for complete SLD with all conductors, OCPD ratings, grounding paths, and NEC labels

SolarPro Engineering

Ray O'Brian Residence Solar
1010 Franklin Street, Pocahontas, IL 62275
Client: Ray O'Brian

E-1

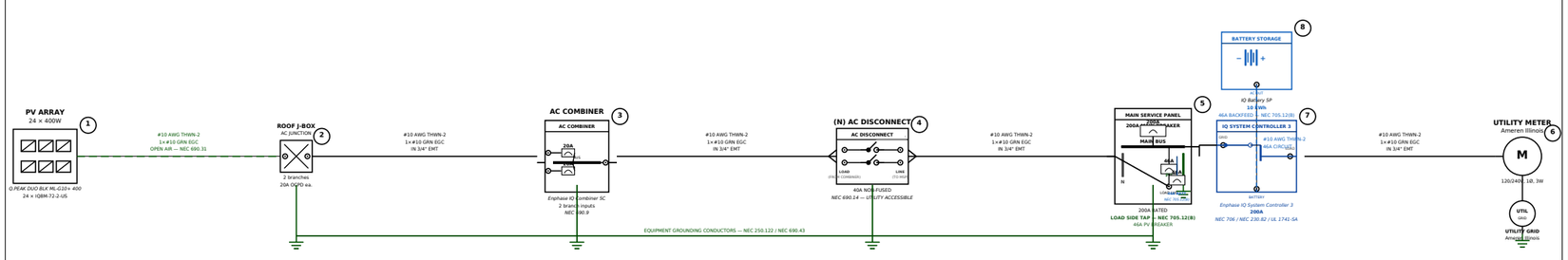
SINGLE-LINE ELECTRICAL DIAGRAM

NEC 2020 · IL · AHJ: Pocahontas / Bond County IL

Designer	SolarPro Engineering
Date	2025-08-12
Sheet	11 of 11
Rev	A
Scale	NTS

SINGLE LINE DIAGRAM — PHOTOVOLTAIC SYSTEM

1010 Franklin Street, Pocahontas, IL 62275 | MICROINVERTER | 24 MODULES | 7.68 kW AC



RAPID SHUTDOWN — NEC 690.12 COMPLIANT — INTEGRATED IN IQ8M MICROINVERTERS

LEGEND

- AC Conductor in Conduit (THWN-2)
- Open Air — PV Wire/THWN-2 (NEC 690.31)
- Equipment Grounding Conductor (EGC)
- Battery AC-Coupled Connection

AC BRANCH CIRCUIT INFO

Topology	MICROINVERTER
Microinverters	24 units
Total DC Power	9.60 kW
AC per Micro	320 W
Branch Circuits	2
Max Micros/Branch	16 (NEC 690.8)
Branch OCPD	20 A
Branch Wire	#10 AWG THWN-2
Feeder Wire	#10 AWG
Feeder Conduit	3/4" EMT
Module Voc	49.6 V
Module Isc	10.5 A

AC SYSTEM CALCULATIONS

AC Output (kW)	7.68 kW
AC Output Amps	32.0 A
AC OCPD (125%)	40 A
AC Wire Gauge	#10 AWG
AC Conduit Type	EMT
Conduit Size	3/4"
Service Voltage	120/240V, 1Ø
Main Panel Rating	200 A
Interconnection	Load Side Tap
NEC Reference	NEC 705.12(B)
PV Breaker	46 A
Batt. Backfeed Bkr	46 A
Total Backfeed	92 A
Bus 120% Limit	240 A
120% Rule	FAIL ✗

EQUIPMENT SCHEDULE

PV Module	Q.PEAK DUO BLK ML-G10+ 400
Module Wattage	400 W
Total Modules	24
Microinverters	24 units
Branch Circuits	2
Inverter Mfr.	Enphase
Inverter Model	IQ8M-72-2-US
Inverter Output	7.68 kW AC
AC Combiner	Enphase IQ Combiner SC
AC Disconnect	40A Non-Fused
Main Panel	200 A
Utility	Ameren Illinois
Interconnection	LOAD_SIDE
Rapid Shutdown	INTEGRATED
Battery Storage	IQ Battery 5P
Battery Capacity	10.0 kWh
Batt. Backfeed	46A — NEC 705.12(B)

CONDUIT & CONDUCTOR SCHEDULE — NEC 310 / NEC CHAPTER 9 TABLE 1

RUN ID	FROM	TO	CONDUCTORS	CONDUIT	FILL %	AMPACITY	OCPD	V-DROP %	LENGTH	PASS
BR-1	PV ARRAY	ROOF J-BOX	#10 AWG THWN-2 + 1×#10 GRN EGC	OPEN AIR	N/A	32A	20A	0.80%	50 FT	✓ PASS
BR-2	ROOF J-BOX	AC COMBINER	#10 AWG THWN-2 + 1×#10 GRN EGC	3/4" EMT	38.2%	32A	20A	1.20%	30 FT	✓ PASS
A-1	AC COMBINER	AC DISCONNECT	#10 AWG THWN-2 + 1×#10 GRN EGC	3/4" EMT	38.2%	32A	40A	1.40%	20 FT	✓ PASS
A-2	AC DISCONNECT	MSP	#10 AWG THWN-2 + 1×#10 GRN EGC	3/4" EMT	38.2%	32A	40A	0.40%	15 FT	✓ PASS
BAT-1	IQ BATTERY 5P	IQ SYS CTRL	#10 AWG THWN-2 + 1×#10 GRN EGC	3/4" EMT	38.0%	46A	46A	0.50%	10 FT	✓ PASS
SVC	MSP	UTILITY METER	SEE UTILITY SERVICE ENTRANCE	SEE UTILITY	—	200A	200A	—	5 FT	✓ PASS

SOLARPRO
ENGINEERING

SINGLE LINE DIAGRAM
PHOTOVOLTAIC SYSTEM
SolarPro V4 — SEED.ORG SLD

PROJECT	Ray O'Brian Residence Solar
CLIENT	Ray O'Brian
ADDRESS	1010 Franklin Street, Pocahont
DESIGNER	SolarPro Engineering
DATE	2025-08-12
DWG NO.	E-1
REVISION	A
SCALE	NOT TO SCALE

SYSTEM SUMMARY

TOPOLOGY	MICROINVERTER
DC SIZE	9.60 kW
AC OUTPUT	7.68 kW
MODULES	24 × 400W
INVERTER	Enphase
MODEL	IQ8M-72-2-US
SERVICE	200A
UTILITY	Ameren Illinois
INTERCONN.	LOAD_SIDE

CODE REFERENCES

- NEC 690 — PV SYSTEMS
- NEC 705 — INTERCONNECTED ELEC.
- NEC 706 — ENERGY STORAGE
- NEC 310 — CONDUCTORS
- NEC 250 — GROUNDING/BONDING
- NEC 358/352 — CONDUIT
- NEC 230 — SERVICES
- IBC / ASCE 7 — STRUCTURAL

REVISIONS

REV	DESCRIPTION	DATE
A	INITIAL ISSUE FOR PERMIT	2025-08-12

ENGINEER SEAL

SolarPro Engineering — 2025-08-12